XUAN-BACH LE

LECTURER
Ho Chi Minh City University of Technology
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EDUCATION

National University of Singapore Jan 2013 – Dec 2017
PhD in Computer Science
Thesis title: *Disjoint Fractional Permissions in Verification: Applications, Systems, and Theory*
Supervisor: Aquinas Hobor Advisor: Anthony W. Lin

National University of Singapore Aug 2007 – Jun 2012
Double Degrees in Computer Science & Mathematics
Specialization: Algorithms and Computation
First-class Honours

EXPERIENCE

Lecturer, Ho Chi Minh City University of Technology, Vietnam Sep 2025 – Now
Safe AI, explainable AI, reinforcement learning, program verification, and reliable software engineering with LLMs and RAG.

Research Fellow, Nanyang Technological University, Singapore July 2023 – Jan 2025
Postdoctoral researcher under Professor Luke Ong. Devised optimality-preserving algorithms to translate LTL and ω-regular objectives for reinforcement learning into limit-average reward machines.

Research Scientist, Singapore Management University, Singapore Jan 2022 – Jan 2023
Formal methods for Program Verification.

Research Fellow, Nanyang Technological University, Singapore May 2019 – Dec 2021
Formalized mathematical reasoning frameworks to verify the correctness of complex systems that feature concurrency, quantum computation, and probabilities. Developed automatic verification tool ReGaSol and decision procedures from these frameworks.

Research Fellow, University of Oxford, UK Aug 2018 – Mar 2019
Algorithmic Verification of String-Manipulating Programs. Investigated the monadic decomposibility problem to decompose complex string constraints into simpler monadic constraints that are solvable. Proved the tight complexity bounds of the decomposibility problem for regular string constraints expressed by DFA and NFA.

Research Associate&Fellow, National University of Singapore, Singapore Sep 2017 – Aug 2018
Developed formal proof system and tool ShareInfer to handle tree share constraints that are used to model permissions in concurrent programs. Investigated methods for symbolic execution to verify programs with loops.

GOOGLE SCHOLAR

https://scholar.google.com.sg/citations?user=FnjHppEAAAAJ

RESEARCH INTERESTS

Logical frameworks and certified decision procedures for program verification; decidability and computational complexity of underlying logics; safe and explainable AI; retrieval-augmented generation; LLM agents for software engineering; program synthesis; reinforcement learning under formal specifications; quantum programming.

PUBLICATIONS

1. Thi-Hong-Cuc Le, Huu-Vu-Phuong Tran, Duc-Thuan Mai, Xuan-Bach Le. *LTM-RAG: Longitudinal Trust Mining for Poison-Resilient Retrieval-Augmented Generation*. In ICDM (Applied Track), 2026.
2. Phat T. Tran-Truong, Son Ha, Quang-Tung Nguyen, Xuan-Bach Le. *From Natural-Language Hypotheses to Verified Data-Mining Discoveries*. In ICDM, 2026.
3. Phat T. Tran-Truong, Xuan-Bach Le, Son Ha. *GraphMemShield: Auditing Cross-Session Leakage in Graph Memories*. In CIKM, 2026.
4. Tien-Anh Nguyen, Thanh-Hai Tran, Xuan-Bach Le. *Test-Time Spectral Adaptation for Image-to-Image Multispectral Retrieval*. In ACIVS, 2026.
5. Thanh-Hai Tran, Xuan-Bach Le. *Latent-Space Prediction Error and Fuzzy Risk-Scaled Recovery for Robotic Manipulation*. In ACIVS, 2026.
6. Van-Truong-Thinh Nguyen, Thanh-Hai Tran, Xuan-Bach Le. *Swin-AQI: Depth-Aware Vision Transformer with Physics-Informed Learning for Particulate Matter Estimation from Ground Imagery*. In ACIVS, 2026.
7. Phat T. Tran-Truong, Xuan-Bach Le. *Distributional Program Analysis: Treating Large Language Models as Program Samplers*. In ASE (NIER Track), 2026.
8. Thi-Hong-Cuc Le, Hoang-Quoc-Bao Hua, Xuan-Bach Le. *GraphLedger: Repository-Level Vulnerability Detection via Graph-Based Compression and Assumption Validation*. In ASE, 2026.
9. Dung-Cam Quang, Xuan-Bach Le, Tho Quan. *Towards Resource-Constrained Event Extraction: A Knowledge-Augmented Framework for Overcoming Challenges in Vietnamese NLP*. In International Journal of Innovative Computing, 16(1), pp. 79–85, 2026.
10. Hieu Le, Surya Hanjaya, Trang Hoang, Xuan-Bach Le. *Examination Room Usage Optimization Via Greedy and Exact Methods: A Vietnamese University Case Study*. In SOMET, 2026.
11. Phat Tuan Tran-Truong, Xuan-Bach Le. *TraceToChain: Auditing LLM-Agent Reliability with Absorbing Markov Chains*. In ISSRE, 2026.
12. Huu-Thanh Phan, Xuan-Bach Le, Tho Quan. *Uncertainty-Aware Forecasting and Inventory Optimization for ATM Cash Management in Vietnam*. In FLINS-ISKE, 2026. Outstanding Student Paper Award.
13. Dung-Cam Quang, Vinh Q. Vo, Xuan-Bach Le, Tho Quan. *The Paradigm Shift in Event Extraction: An In-Depth Overview with Large Language Models*. In IEA/AIE, 2026.
14. Hoang-Quoc-Bao Hua, Xuan-Bach Le. *Emerging Trends and Challenges Toward LLM Agents in Static Analysis*. In IEA/AIE, 2026. Nominated for Best Paper Award.
15. Thi-Hong-Cuc Le, Xuan-Bach Le. *From Black-Box to Glass-Box: Explainable AI for Applied Intelligent Software Systems Across the Software Development Life Cycle*. In IEA/AIE, 2026. Best Paper Award.
16. Thanh-Hai Tran, Xuan-Bach Le. *Industrial Visual Anomaly Detection in Robotics: Methods, Datasets, and Deployable System Architectures*. In IEA/AIE, 2026.
17. Phong Chung, Kha Le-Minh, Xuan-Bach Le, Tho Quan. *VSLIM: A Vietnamese Explicit Slot-Intent Mapping for Joint Multi-Intent Detection and Slot Filling*. In ACIIDS, 2026.
18. Khoa Phan, Xuan-Bach Le, Tho Quan. *SBV-LawGraph: A Hybrid RAG Approach Integrating Knowledge Graph for the State Bank of Vietnam Legal Documents*. In ACIIDS, 2026. Oral presentation.

19. Xuan-Bach Le, Dominik Wagner, Leon Witzman, Alexander Rabinovich, Luke Ong. *Reinforcement Learning with LTL and ω-Regular Objectives via Optimality-Preserving Translation to Average Rewards.* In NeurIPS, 2024.
20. Xuan-Bach Le, Shang-Wei Lin, Sun Jun, David Sanán. *A Quantum Interpretation of Separating Conjunction for Local Reasoning of Quantum Programs Based on Separation Logic.* In POPL, 2022.
21. Xuan-Bach Le, David Sanán, Sun Jun, Shang-Wei Lin. *Automatic Verification of Multi-threaded Programs by Inference of Rely-Guarantee Specifications.* In ICECCS, 2020.
22. Pablo Barceló, Chih-Duo Hong, Xuan-Bach Le, Anthony W. Lin, Reino Niskanen (alphabetical order). *Monadic Decomposability of Regular Relations.* In ICALP, 2019.
23. Xuan-Bach Le, Aquinas Hobor, Anthony W. Lin. *Complexity Analysis of Tree Share Structure.* In APLAS, 2018.
24. Xuan-Bach Le, Aquinas Hobor. *Logical Reasoning for Disjoint Permissions.* In ESOP, 2018.
25. Xuan-Bach Le, Thanh-Toan Nguyen, Wei-Ngan Chin, Aquinas Hobor. *A Certified Decision Procedure for Tree Shares.* In ICFEM, 2017.
26. Xuan-Bach Le, Aquinas Hobor, Anthony W. Lin. *Decidability and Complexity of Tree Share Formulas.* In FSTTCS, 2016.
27. Xuan-Bach Le, Cristian Gherghina, Aquinas Hobor. *Decision Procedures Over Sophisticated Fractional Permissions.* In APLAS, 2012.

TEACHING EXPERIENCE AND SERVICES

Lecturer, Ho Chi Minh City University of Technology, Vietnam 2025
– 2026 CO1005 (Introduction to Computing, Semester 251, 2025), CO2001 (Professional Skills for Engineers, Semester 252, 2026), CO2039 (Advanced Programming, Semester 252, 2026)

Teaching Assistant and Lab Assistant, National University of Singapore, Singapore 2009 – 2015
CS1231 (Discrete Maths, S1/11-12, S1/13-14), CS1010 (Programming Methodology, S1/10-11), CS1102/1020 (Data Structures I, S2/09-10, S2/10-11), CS3234 (Logic & Formal Systems, S1/10-11, S2/12-13)

Grader, Yale-NUS, Singapore Jan 2017 – May 2017

YCC1212 (Introduction to Computer Science, S2/16-17), YCC2202 (Data Structures & Algorithm, S2/16-17)

Professional Services. Program Committee Member of APLAS 2022. External reviewer of APLAS 2016, ICALP 2017, PLDI 2018, LICS 2018, APLAS 2018, VMCAI 2019.

SKILLS

Programming Languages and Frameworks

AI/ML:	PyTorch, SVM, probabilistic forecasting, RAG, LLM-based software analysis
Solvers and proof assistants:	Coq, Z3, CVC3
Programming:	Python, Java, C++, OCaml
Web:	JavaScript, PHP, HTML, MYSQL

Languages
Vietnamese: Mothertongue, English: Fluent, French: Basic Knowledge

AWARDS AND SCHOLARSHIPS

President Graduate Fellowship 2013 – 2017
ASTAR Pre-graduate Scholarship 2007–2012
NUS Donated Scholarship. Dean List Award 2007, 2008, 2010. Certificate of Special Program in Computing.

REFEREES

Aquinas Hobor, Associate Professor in Computer Science, University College London, UK
- Email: `a.hobor@ucl.ac.uk` • Tel: (+44) 777-846-3201 (UK)
- Website: https://profiles.ucl.ac.uk/87909-aquinas-hobor

James Brotherston, Associate Professor in Computer Science, University College London, UK
- Email: `J.Brotherston@ucl.ac.uk` • Tel: (+44) 787-124-0131 (UK)
- Website: http://www0.cs.ucl.ac.uk/staff/J.Brotherston/

Alexander Rabinovich, Professor in Computer Science, Tel Aviv University, Israel
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- Website: http://www.math.tau.ac.il/ rabinoa/

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